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i

ABSTRACT

Phishing websites pose an ongoing threat by impersonating legitimate services to extract user credentials and sensitive information. Despite advancements in cybersecurity, accurately detecting these malicious domains in real-time remains a critical challenge due to high computational latency and the rapid emergence of "zero-day" phishing sites. This project develops a highly optimized, URL-based automated phishing detection framework integrating Rough Set Theory-based Hybrid Feature Selection (RSTHFS) with an innovative Meta-Learning Ensemble.

The approach focuses on safe, lexical, and host-based URL feature extraction, prioritizing an optimized "minimal reduct" of features identified through RSTHFS. This optimization reduces computational overhead significantly. The system processes these optimized features through an ensemble architecture leveraging machine learning classifiers aggregated by a meta-classifier to enhance predictive stability and achieve high detection accuracy.

To ensure practical applicability, the system is engineered as a decoupled architecture featuring a Python-based FastAPI backend inference engine and a React-based frontend. Furthermore, the framework incorporates Explainable AI (XAI) through SHAP (SHapley Additive

exPlanations) values to provide granular transparency, moving beyond "black-box" predictions by delivering specific, feature-based justifications for every classified site. The final deliverable is a comprehensive, near-real-time URL classification system emphasizing high detection performance, reduced computational cost, safety-by-design, and user-interpretable alerts.

Keywords: Phishing Detection, Ensemble Learning, SHAP, Explainable AI (XAI), RSTHFS, URL Security, Feature Optimization, FastAPI, React.

ii

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